MATH3603 Probability Theory
Solution to Assignment 1

Solution to Problem 1:

Proof. Note that we already prove for ANY two events E_1, E_2,

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2). \tag{1}$$

By induction, assume that we already have for ANY n-1 events $E_1, E_2, \cdots, E_{n-1}$,

$$P(E_1 \cup E_2 \cup \cdots \cup E_{n-1}) = \sum_{i \le n-1} P(E_i) - \sum_{i<j \le n-1} P(E_i \cap E_j) + \cdots + (-1)^n P(E_1 \cap E_2 \cap \cdots \cap E_{n-1}). \tag{2}$$

Applying formula 1, we have

$$P((E_1 \cup E_2 \cup \cdots \cup E_{n-1}) \cup E_n) = P(E_1 \cup E_2 \cup \cdots \cup E_{n-1}) + P(E_n) - P((E_1 \cup E_2 \cup \cdots \cup E_{n-1}) \cap E_n)$$

Applying formula 2 to $P(E_1 \cup E_2 \cup \cdots \cup E_{n-1})$ and $P((E_1 \cup E_2 \cup \cdots \cup E_{n-1}) \cap E_n) = P((E_1 \cap E_n) \cup (E_2 \cap E_n) \cup \cdots \cup (E_{n-1} \cap E_n))$, we obtain

$$P(E_1 \cup E_2 \cup \cdots \cup E_{n-1} \cup E_n) = \sum_{i \le n-1} P(E_i) - \sum_{i<j \le n-1} P(E_i \cap E_j) + \cdots + (-1)^n P(E_1 \cap E_2 \cdots \cap E_{n-1}) + P(E_n)$$

$$-\Big(\sum_{i \le n-1} P(E_i \cap E_n) - \sum_{i<j \le n-1} P(E_i \cap E_j \cap E_n) + \cdots + (-1)^n P(E_1 \cap E_2 \cap \cdots \cap E_n)\Big).$$

Collecting terms, we finally obtain

$$P(E_1 \cup E_2 \cup \cdots \cup E_n) = \sum_i P(E_i) - \sum_{i<j} P(E_i \cap E_j) + \sum_{i<j<k} P(E_i \cap E_j \cap E_k)$$

$$- \sum_{i<j<k<l} P(E_i \cap E_j \cap E_k \cap E_l) + \cdots + (-1)^{n+1} P(E_1 \cap E_2 \cap \cdots \cap E_n).$$

□

Solution to Problem 2:

Proof. Notice that for any i with $2 \le i \le n$, we have

$$P(E_i | E_1 \cap E_2 \cap \cdots \cap E_{i-1}) = \frac{P(E_1 \cap E_2 \cap \cdots \cap E_i)}{P(E_1 \cap E_2 \cap \cdots \cap E_{i-1})}.$$

Plug all these $n-1$ equalities into the right hand side of the equality we are asked to prove, we obtain

$$P(E_1) p(E_2|E_1) P(E_3|E_1 \cap E_2) \cdots P(E_n|E_1 \cap \cdots \cap E_{n-1})$$

$$= P(E_1) \frac{P(E_1 \cap E_2)}{P(E_1)} \cdots \frac{P(E_1 \cap E_2 \cap \cdots \cap E_n)}{P(E_1 \cap E_2 \cap \cdots \cap E_{n-1})} = p(E_1 \cap E_2 \cap \cdots \cap E_n).$$

□

Solution to Problem 3:

Proof. Let $F_1 = E_1$ and $F_i = E_i \cap (\cap_{j=1}^{i-1} E_j^c)$ for any i with $2 \leq i \leq n$. It is straight forward to prove that $F_i \cap F_j = \phi$ for $i \neq j$, and $\cup_{i=1}^n F_i = \cup_{i=1}^n E_i$. So

$$P(\cup_{i=1}^n E_i) = P(\cup_{i=1}^n F_i) = \sum_{i=1}^n P(F_j) \leq \sum_{i=1}^n P(E_j),$$

where the last ineqaulity follows from $F_i \subset E_i$ for any i. □

Solution to Problem 4:

(i) Let A be the event that the randomly chosen ruler is defective. Let E_1 be the event that the ruler is chosen from factory I. Let E_2 be the event that the ruler is chosen from factory II. By the law of total probability,

$$\begin{aligned} P(A) &= P(A|E_1)P(E_1) + P(A|E_2)P(E_2) \\ &= \frac{1}{5} \cdot \frac{2}{3} + \frac{1}{20} \cdot \frac{1}{3} \\ &= \frac{3}{20}. \end{aligned}$$

Then

$$P(\{\text{The randomly chosen ruler is satisfactory}\}) = 1 - P(A) = \frac{17}{20}.$$

(ii)

$$\begin{aligned} P(E_1|A) &= \frac{P(A \cap E_1)}{P(A)} \\ &= \frac{P(A|E_1)P(E_1)}{P(A)} \\ &= \frac{\frac{1}{5} \cdot \frac{2}{3}}{\frac{3}{20}} \\ &= \frac{8}{9}. \end{aligned}$$

Solution to Problem 5:

(i) Notice that

$$P(A) = P(A \cap B) + P(A \cap B^c),$$

then

$$\begin{aligned} P(A \cap B^c) &= P(A) - P(A \cap B) \\ &= P(A) - P(A)P(B) \\ &= P(A)(1 - P(B)) \\ &= P(A)P(B^c). \end{aligned}$$

(ii) By the independence of A, B, C,

$$\begin{aligned} P(A \cap (B \cap C)) &= P(A)P(B)P(C) \\ &= P(A)(P(B)P(C)) \\ &= P(A)P(B \cap C). \end{aligned}$$

(iii)

$$\begin{aligned} P(A \cap (B \cup C)) &= P((A \cap B) \cup (A \cap C)) \\ &= P(A \cap B) + P(A \cap C) - P(A \cap B \cap C) \\ &= P(A)P(B) + P(A)P(C) - P(A)P(B)P(C) \\ &= P(A)(P(B) + P(C) - P(B \cap C)) \\ &= P(A)P(B \cup C). \end{aligned}$$